

1711 – HyFleet

Functional Design Specification

[FDS]

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1. Introduction

The purpose of this Functional Design Specification (FDS) is to provide a common technical reference for the HyFleet system. It sets out the intended functions, scope and integration requirements so that internal engineering teams have a consistent basis for design, implementation and review. The FDS is not intended as a certification or customer-facing document, but as an internal alignment tool across mechanical, electrical, process, and control disciplines.

Scope :

This specification covers the HyFleet core system: inlet and recirculation, compression, smart bank storage management, cooling, and site interfaces. HyFlow dispensers are treated as separate products with their own FDS, but they are referenced where necessary to define how HyFleet provides supply pressure and communicates with them. The document also considers deployment of multiple HyFleet units operating on the same site, coordinating compression, storage utilisation, and support for up to two HyFlow dispensers per unit.

Intended Audience :

The audience for this document is FCSL internal engineers working across mechanical, electrical, controls, and process teams. It provides a high-level overview of function and scope, with detailed behaviour, sequencing, and software architecture addressed in the Software Design Specification (SDS) and subsequent documents.

Document Relationships :

This FDS should be read in conjunction with the following project documents:

- Software Design Specification (SDS) – details control sequences, state logic, and software architecture.
- ATEX Register – records hazardous area classification, equipment certifications, and inspection requirements.
- Certification Pack – collates CE/UKCA compliance evidence against applicable standards.

2. System Overview

HyFleet is a hydrogen refuelling system designed to compress hydrogen gas from a variety of sources, manage on-site storage, and supply pressurised hydrogen to paired HyFlow dispensers. Each HyFleet unit integrates compression, smart bank storage management, cooling, and control to provide reliable, efficient refuelling capability.

Subsystems :

- **Inlet & Recirculation :**

Receives hydrogen from electrolyzers, tube trailers, or other sources at 20–500 bar. Functions include manual isolation, filtration, pressure regulation, and routing of gas.

H₂ can be routed from either H₂ inlet or via smart bank into either compression stages or directly to onsite storage or dispensing as required. **Note :** input H₂ line and both feeds to compression skid feed via double block and bleed valve to allow safe manual isolation and venting of sub-system without purging whole system.

Compression :

A dual-stage Haskel booster skid powered by a Rexroth hydraulic power unit (HPU). The system can be configured in two ways: operating in series to raise pressure from ~20 bar inlet up to ~900+ bar, or as two independent stages (low and high) to optimise energy usage and throughput depending on inlet conditions. Major items include the Haskel compressors, hydraulic pumps, accumulator, control valves, and instrumentation for monitoring and protection.

- **Smart Bank Manifold :**

A valve manifold designed to optimise system performance and gas flow. It supports four storage tanks, each protected by a 990 bar PRV. The manifold incorporates a two-split 3×4 solenoid matrix that provides routing for: four tanks, two inputs from the Haskel skid (stage 1 and stage 2), one bypass from the inlet, two HyFlow outlets, and a recirculation line back to the inlet. It provides cascade management and flexibility to operate in dual-stage or series compression configurations.

Note : input to onsite tanks and both feeds from compression fed via double block and bleed valve to allow safe manual isolation and venting of sub-system without purging whole system.

Note : Supports inlet to allow direct cascade fill directly to dispensing.

- **Cooling System :**

Removes heat from the hydraulic power unit (HPU) and compressor jackets to ensure efficient and stable operation. Major items include a dedicated chiller, circulation pumps, heat exchangers, and temperature sensors. The cooling loop is sized for continuous duty operation, protecting both hydraulic and gas compression equipment. Consists of 3 primary lines:

- HPU via heat exchange.
- Stage 1 cooling line, including pre and post tube in tube heat exchange and cooling jacket on booster.
- Stage 2 cooling line. including post tube in tube heat exchange and cooling jacket on booster.

- **Control System :**

A Beckhoff PLC with EtherCAT IO coordinates all subsystems. Functions include execution of system state management, cascade logic, and interlocks. It manages communication with HyFlow dispensers, HySentinel site safety, and HyCloud monitoring. Local HMIs provide operator access and service tools. Major items include PLC CPU, remote IO terminals, fibre/Ethernet comms modules, and HMI panels.

All H2, fire sensors are fed via a Gas Master 4 (monitored using RS-485) which along with E-Stop and other safety systems are hardwired back to central PLC and are controlled using a Sil.3 safety rated controller. Also present are dual channel volt free contact to site connect to site-wide safety system.

- **Miscellaneous Systems :**

Provides supporting utilities required for safe and reliable operation. Includes CDA (compressed dry air) preparation for pneumatic actuators, auxiliary power distribution, lighting and heating within enclosures, and instrumentation support systems. Major items include air dryers and filters, distribution manifolds, auxiliary control relays, and ancillary monitoring devices.

Interfaces

- Tube trailers / electrolyser inputs for hydrogen supply.
- External storage banks for high-pressure hydrogen storage.
- Up to 2 HyFlow dispensers for vehicle refuelling.
- Site power and communications, including integration with site-wide safety via HySentinel and secure remote monitoring via Tosibox.

3. Applicable Standards & References

HyFleet must comply with a range of international and European standards to ensure safety, reliability, and certification. At a high level these can be grouped as follows:

- **CE/UKCA Conformity Standards** – HyFleet is covered under the Machinery Directive and Pressure Equipment Directive (PED) for its drives, compression, and storage elements. The Low Voltage Directive and Simple PED are not applicable. Compliance also includes the EMC Directive and ATEX Directive.
- **Hydrogen Compression & Refuelling Standards** – Key sector-specific standards governing hydrogen fuelling stations, compression, storage, and dispensing. This includes ISO 19880 for fuelling stations and SAE J2601 fuelling protocols, with further references captured in Annex E (ISO 17268 connectors, ISO 14687 fuel quality, BS EN 17127, and relevant BCGA Codes of Practice).
- **Functional Safety & Cybersecurity Standards** – Compliance with IEC 61511 for functional safety of process systems, IEC 62443 for industrial automation security, and ISO/IEC 27001 for information security management. The detailed safety philosophy and implementation are described in the SDS.

Reference documents for this specification include the HyFleet Intent, P&ID drawings, requirements matrix (RVM), and associated design documents.

The full standards list with references and detailed compliance mapping will be provided in Annex E and in the separate Certification Pack. These standards and references form the framework against which HyFleet will be designed, tested, and certified.

4. Functional Requirements

HyFleet must meet the following high-level functional requirements, grouped by category.

Performance

- Inlet operating range: 20–500 bar hydrogen supply.
- Outlet pressure range: 200–900 bar for storage and dispensing.
- Nominal throughput: ~35 kg/hr (exact per day throughput TBD, dependent on duty cycle/operating hours).
- Supports vehicle refuelling at 350 bar and 700 bar protocols (SAE J2601 compliant).

Operating Envelope

- Designed for continuous operation in typical outdoor UK/EU climates.
- Ambient operating temperature range: -20 °C to +40 °C (extended ranges possible with site-specific design).
- Capable of concurrent compression and dispensing.

Reliability & Availability

- Dual-stage compression allows continued operation in degraded mode (one stage available).
- Smart Bank manifold enables flexible routing and cascade management.
- Designed for availability >95% across operating year, subject to scheduled maintenance.

Maintainability & Support

- Modular design for ease of access and replacement of subsystems (compressors, valves, sensors).
- Access detection interlocks on all service panels and doors.
- Remote diagnostics, logging, and software updates enabled via secure Tosibox link.

Safety & Compliance

- Fail-safe design: NC valves (exception: Haskel skid NO valves venting on CDA loss/E-Stop).
- Over-pressure managed by software limits, regulators, and PRVs.
- Compliance with CE/UKCA directives and hydrogen refuelling standards (see Section 3).
- Hazardous area compliance in line with ATEX (see Section 8).

5. System Architecture

HyFleet is composed of several integrated subsystems, each responsible for a distinct function. Together they form the overall architecture that delivers compression, storage, and dispensing capability.

5.1 Inlet & Recirculation

The inlet section consists of a compact 3×2 smart valve matrix connected to a single hydrogen inlet and a recirculation line. It can feed either the first stage, the second stage, or bypass compression. Parallel operation is supported with the limitation that a single input can only be connected to a single output at any time (e.g., inlet gas may feed stage 1 while recirculating gas to stage 2 independently). The outlet to the Haskell skid is protected by a double block-and-bleed arrangement, allowing safe service and maintenance of the system.

5.2 Compression

A dual-stage Haskell booster skid powered by a Rexroth hydraulic power unit (HPU). Can operate in series (20 bar up to ~900+ bar) or as two independent stages (low and high) depending on inlet conditions. Major components include compressors, hydraulic pumps, accumulator, valves, and instrumentation.

5.3 Smart Bank Manifold

Optimises system performance and gas flow. Supports four storage tanks, system overpressure protection using PRV, and incorporates a two-split 3×4 solenoid matrix. Provides routing for tanks, two compressor inputs, bypass line, two HyFlow outlets, and recirculation. Enables cascade control and flexible operating modes.

5.4 Cooling System

A The cooling system consists of a Parker HyperChill 120E chiller (located external to HyFleet) and a Zone 2 water manifold distributing coolant across three dedicated circuits:

1. Hydraulic Power Unit (HPU)
2. Stage 1 Booster and its intercooler
3. Stage 2 Booster and its intercooler

The chiller communicates via RS-485, supporting alarms, setpoint adjustment, run/stop commands, and temperature feedback. The manifold includes:

- Isolation valves on supply lines
- Manual flow restrictors for commissioning and load balancing
- Return temperature monitoring and check valves
- Dedicated pressure transducer on the HPU circuit
- Pressure transducers on Stage 1 and Stage 2 coolant lines
- Needle valve between supply and return for recirculation and to prevent dead-head conditions
- Drain and fill ports for coolant maintenance

5.5 Control System

The main control and electrical equipment are housed in a non-hazardous zone. This includes AC and DC power distribution, drives, PLC, IPC, external comms (Tosibox), internal managed switch, and IO for solenoids.

The IO architecture follows a distributed EtherCAT design with islands of IO located close to field devices (primarily in Zone 2) to minimise cable runs.

Operator and engineering interfaces include a local WebHMI for engineering access, an HMI panel for local operation, and a stack light mounted on the control panel for visual status indication. RFID is used for engineering-level access control.

5.6 Miscellaneous System

The main miscellaneous systems provide essential support utilities:

- **Compressed Dry Air (CDA):** An air preparation unit comprising isolation valve, filter, regulator, and SIL 3-rated dump valve. Supplies a 10-station (20-output) dual NC solenoid bank for actuation of pneumatic isolation valves.
- **Gas/Fire Monitoring:** A GasMaster 4 panel connected to three hydrogen sensors and up to three fire/smoke detectors, providing detection signals into the safety system. Hardwired dual channel safety back to safety controller, remote control via RS-485 (including remote reset and values).
- **Enclosure Systems:** Lighting and heating for safe access and winterisation. Zone 2 fans for ventilation to maintain hazardous area compliance.

6. Control & Operation

6.1 Operational States

HyFleet operates under four primary states:

- Initialise – System boot, integrity checks, and preparation for operation.
- Manual – Operator-driven mode for maintenance, commissioning, and troubleshooting.
- Auto – Normal operating mode, managing compression, storage, and refuelling sequences concurrently.
- Lockout – Safe state enforced on detection of a critical fault, E-Stop, hydrogen leak, or fire. Requires manual reset after resolution.

Each major subsystem transitions in line with the overall state, enabling compression and refuelling to run concurrently where permitted.

6.2 Control Software

HyFleet builds upon the HyCube V3 software framework, reusing the same base library. The control software is structured into the following layers and managers:

- Orchestration Layer – Top-level coordination of subsystems and overall sequencing.
- IO Manager – Handles configuration-driven IO mapping, enabling hot-swapping of IO modules.
- Device Manager – Executes device-level logic for pumps, valves, compressors, and sensors.
- HMI Manager – Provides interface to external HMIs and engineering tools.
- Subsystem Controllers – Independent modules for compression, gas management, and dispensing, each coordinated by the orchestration layer.

Detailed sequencing and logic are documented in the SDS.

6.3 Network Architecture

The HyFleet network is divided into two distinct domains:

- **Operational Technology (OT):** Beckhoff PLC running TwinCAT 4016 with EtherCAT IO for high-speed, deterministic control. Drives communicate with the PLC via Profinet. This network handles all safety-critical and time-sensitive operations.
- **Information Technology (IT):** Provides data logging, ROS2 integration, and remote support. Communicates over Ethernet/IP through managed industrial switches. Interfaces include HyFlow (treated as an extension of HyFleet), while HySentinel (site-wide safety) and HyCloud (remote monitoring) are external systems connected via secure interfaces.

Security Measures :

All external connections are routed through Tosibox to provide a secure VPN. Internal traffic is segmented and protected by security-managed switches, ensuring robust cybersecurity separation between IT and OT domains.

6.4 Multi-unit Deployments

- Multiple HyFleet units can operate in parallel, coordinating compression and storage utilisation.
- Shared smart bank logic ensures balanced use of available storage capacity.
- Control logic prevents conflicts when multiple HyFlows request refuelling simultaneously.
- Multi-input and multi-bank configurations are managed upstream through HyLink panels, which provide routing and isolation before connection into HyFleet.

6.5 Reporting & Support

- Continuous logging of operational data, power consumption, and alarms.
- Remote diagnostics and updates available via Tosibox-secured connection.
- Aggregated reporting provided for customer interfaces and internal maintenance.
- Live compression seal monitoring along with hour run data to allow for service/maintenance activities.

7. Safety Functions

HyFleet is designed around a fail-safe safety philosophy: under power loss or emergency stop conditions, the system moves to a safe state. All solenoid valves are normally-closed (NC), including pneumatic isolation valves, to contain hydrogen. The exception is the Haskel skid, which uses normally-open (NO) valves that vent tanks on loss of CDA or E-Stop, ensuring depressurisation of the compression train. Safety logic is implemented in the Beckhoff PLC using Beckhoff Safety IO.

7.1 Detection

- Hydrogen detectors located within enclosures and process areas.
- Fire/smoke detectors within key enclosures.
- Both detector types connect to a GasMaster 4 panel, which provides hardwired signals into the safety system. Communication to the PLC is via RS-485 for information only.
- Pressure and temperature sensors for early detection of abnormal conditions.
- Access detection on all doors and removable panels, interlocked with the control system.
- Final detector locations will be confirmed during the Hazardous Area Classification (HAC) assessment.

7.2 Protection

- Three levels of over-pressure protection:
 1. Control limits via software using PT/TT instrumentation.
 2. Pressure regulators for sensitive areas or where pressure may be generated.
 3. Pressure Relief Valves (PRVs) as final backup to maintain system integrity.
- Thermal Pressure Relief Devices (TPRDs) for additional equipment protection.
- Intrinsic Safety (IS) barriers for sensors and devices located in hazardous zones.
- Segregation of hydrogen and electrical systems in line with ATEX and IEC standards.

7.3 Isolation

- Pneumatic isolation valves (NC) to contain hydrogen on fault or power loss.
- Double block and bleed isolation valves at key interfaces (inlet, storage, dispenser connections).
- Electrical isolation via main disconnects and interlocked panel doors.
- Segregation of hydrogen and electrical systems in line with ATEX and IEC standards.

7.4 Response

- Emergency Stop (E-Stop) pushbuttons located at operator points and key access areas (final locations defined in HAC assessment).
- Automatic lockout state triggered by hydrogen detection, fire, fan failure, critical faults, or manual E-Stop.
- Controlled venting through dedicated vent stacks and Haskel skid vent path.

7.5 Applicable Standards

Safety functions are designed in line with:

- IEC 61511 (Functional Safety)
- ATEX Directive 2014/34/EU
- PED 2014/68/EU
- BCGA Codes of Practice for hydrogen systems

7.6 Summary

Category	Key Elements
Detection	H2 detectors, Fire/smoke detectors → GasMaster 4, PT/TT sensors, Access detection
Protection	3-level over-pressure (software, regulators, PRVs), TPRDs, IS barriers, ATEX segregation
Isolation	NC pneumatic valves, Manual isolation valves, Electrical disconnects/interlocked doors
Response	E-Stops, Automatic lockout (triggered by leak, fire, fan failure, faults), Controlled venting via vent stacks/Haskel skid

8. ATEX Considerations

HyFleet is designed and assessed in accordance with ATEX and IECEx principles. The hazardous area classification for the system has been defined as follows:

- **Zone 2** – Booster section of the HyFleet compression skid, ventilated by two large fans. Each fan is sized so that a single fan can maintain adequate airflow to meet Zone 2 requirements. Fans are monitored via current sensors and differential pressure switches to verify operation. Failure of a single fan is treated as an Emergency Stop condition, driving the system to lockout.
- **Non-Hazardous Zones** – Control system cabinets and the hydraulic power unit (HPU) are located outside of hazardous zones.
- **Zone 1 (HyFlow)** – HyFlow dispensers are classified as Zone 1 due to lack of forced ventilation.

Equipment Certification

All electrical and mechanical equipment installed in hazardous areas is certified to the relevant Ex standards. This includes valves, sensors, transmitters, cables, and glands.

Cabling & Segregation

- Intrinsic Safety (IS) barriers provided for instrumentation in hazardous zones.
- Blue sleeving and labelling applied for IS circuits.
- Roxtec ATEX-rated glands used for transitions between hazardous and non-hazardous zones.
- Cable routing and segregation in line with EN 60079 series.

Inspection & Maintenance

- Regular inspection and maintenance of hazardous area equipment in accordance with ATEX/IECEx requirements.
- Records maintained in the ATEX Register deliverable.
- Detailed zoning and equipment layouts will be defined in the HAC drawing.

Applicable Standards

- ATEX provisions are applied in accordance with:
- ATEX Directive 2014/34/EU
- EN 60079 series (equipment, installation, inspection)
- IECEx scheme requirements

All compliance evidence will be collated in the ATEX Register, with this section providing the overarching principles.

9. Abnormal Situations

HyFleet has been designed to handle a range of abnormal situations. These are grouped by type, with the expected system response summarised below.

Loss of Utilities

- **Power Loss** – On loss of mains power the system transitions to a safe lockout state. A 24 V UPS maintains the PLC, safety system, and communications to ensure controlled shutdown, logging, and remote monitoring. Compression and dispensing stop, and valves return to fail-safe positions as described in Section 7.
- **Loss of CDA** – Pneumatic isolation valves close on loss of compressed dry air, containing hydrogen. Haskel skid NO valves vent to depressurise the compression train.

Equipment Failures

- **Compressor Failure** – Fault detection triggers isolation of the affected stage. System may continue in degraded mode if one compression stage remains available.
- **Chiller Failure** – Over-temperature conditions initiate controlled shutdown of compression. Dispensing may continue for a limited period depending on stored capacity.

Sensor Failures

- Detected through diagnostics and plausibility checks. Faulted sensors are isolated in control logic and trigger alarms. Critical measurement failures result in transition to lockout until maintenance is completed.

Safety Events

- **Hydrogen Leak Detection / Fire / Emergency Stop** – All treated identically. All solenoid valves close, the system enters lockout, and the Haskel skid NO valves vent to depressurise the compression train.
- **Over-pressure** – Three levels of protection are provided: (1) control via software pressure limits; (2) pressure regulators where over-pressure may be generated; and (3) PRVs to maintain system integrity. All process lines are equipped with pressure and temperature monitoring. Hydrogen boosters additionally include temperature monitoring, end-of-travel switches, and TPRDs for protection.

10. Power Management

HyFleet includes a built-in scheduler and two selectable operating modes to manage site power constraints and availability of high-pressure gas. Default behaviour is unrestricted power, with the option to apply a site power cap down to a minimum of 150 kW when required.

Operating Modes

- Eco Mode (aim: kg per kWh) – Optimises energy usage to maximise hydrogen delivered per unit energy. Prefers single-stage operation where feasible, minimises recompression, and smooths demand peaks. Modes are selectable via the HMI or supervisory system.
- Performance Mode (aim: availability of high-pressure gas) – Prioritises maintaining bank pressures for dispensing. Allows higher instantaneous demand to ensure HyFlow availability. Modes are selectable via the HMI or supervisory system.

Scheduler & Policies

- Time-of-day/week scheduling to switch modes (e.g., Efficiency off-peak, Throughput at peak depot hours).
- Policy-driven decisions for when to start/stop compression based on bank pressures and forecast demand.
- Operator or supervisory (HyCloud) override permitted within configured limits.

Site Power Constraint Handling

When a cap is applied (≥ 150 kW), HyFleet adapts by:

- Limiting compression to one stage at a time where needed (e.g., fill to ~ 300 bar, recirculate to lift a second bank to ~ 500 bar).
- Reducing strokes-per-minute to lower instantaneous hydraulic load.
- Sequencing cooling loads and auxiliaries to avoid coincident peaks.
- Preferring storage bank selection that minimises recompression energy.

Energy Feedback & Instrumentation

- Energy/consumption telemetry required for all devices > 1 kW (HPU, chiller, fans, auxiliaries).
- Aggregated power/energy metrics exposed for local HMI and HyCloud reporting.
- Alarms on sustained exceedance of configured power limits.

Avenues to Reduce Power (guidance)

- Operate in Eco Mode during low demand; delay bank topping until needed.

- Switch to Performance Mode during peak demand to ensure availability.
- Optimise cascade logic to avoid unnecessary equalisation losses.
- Prefer recirculation strategies over full recompression when inlet conditions allow.
- Curtail non-essential auxiliaries during capped periods.

NOTE : Future Development: Power management is a key R&D area. FCSL is extending control to ROS2 and evaluating model-based and AI optimisation to forecast demand, schedule compression, and dynamically tune setpoints for energy/performance balance.

11. Site Integration

HyFleet acts as the beating heart of a hydrogen refuelling site, providing compression and storage management that enables reliable hydrogen dispensing through paired HyFlow dispensers. It integrates into site infrastructure through defined electrical, utility, process, and safety interfaces.

Electrical Supplies

- Main supplies: 2 × 125 A 3Ph+E and 1 × 32 A 3Ph+N+E combined line.
- External chiller (site-provided): 1 × 125 A 3P+E dedicated supply.

Utilities

- Compressed Dry Air (CDA): Clean, dry air > 8 bar, sized for pneumatic actuators and instrumentation.

Venting

- Vent stack integrated within HyFleet for controlled hydrogen releases (see Sections 7 & 9 for safety behaviour).

Process Interfaces

- Inlet: Single hydrogen input connection to HyFleet. If multiple inputs are required, this is achieved via HyLink interface panels upstream.
- Storage: Four on-skid tanks. Banks can be split or expanded via HyLink panels where site requires multiple banks.
- Dispense outlets: 2 dedicated outputs for HyFlow dispensers (per HyFlow FDS).

Safety & Comms Integration

- Integration to HySentinel for site-wide safety circuit signalling.
- Secure remote monitoring/logging via Tosibox; internal network locked down to cybersecurity best practice.

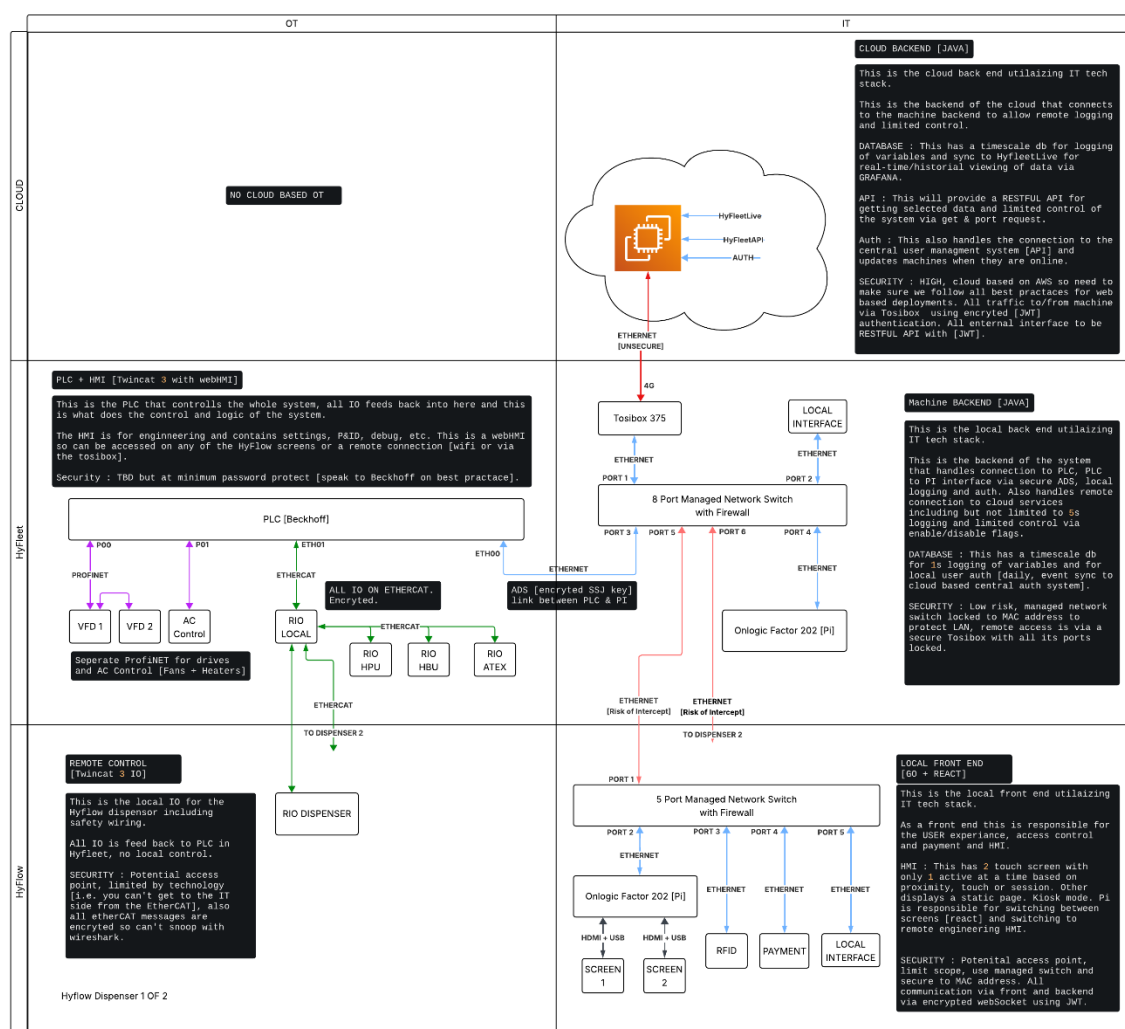
12. Annex A – Abbreviations

Abbreviation	Expansion
CDA	Compressed Dry Air
HPU	Hydraulic Power Unit
PLC	Programmable Logic Controller
IO	Input/Output
HMI	Human Machine Interface
PRV	Pressure Relief Valve
TPRD	Thermal Pressure Relief Device
SDS	Software Design Specification
FDS	Functional Design Specification
ATEX	Atmospheres Explosible
PED	Pressure Equipment Directive
EMC	Electromagnetic Compatibility
VPN	Virtual Private Network
RS-485	Recommended Standard 485
ROS2	Robot Operating System 2
BCGA	British Compressed Gases Association
IS	Intrinsic Safety
UPS	Uninterruptible Power Supply
IPC	Industrial PC
IT	Information Technology
OT	Operational Technology
Eco Mode	Energy-efficient power mode
Performance Mode	High-throughput power mode

13. Annex B – Design Data

- Inlet: 20–500 bar
- Outlet/storage: 200–950 bar
- PRV: 990 bar
- Throughput: ~35 kg/hr
- Refuelling: 350/700 bar SAE J2601
- Supplies: 2 × 125 A 3Ph+E + 1 × 32 A 3Ph+N+E, 1 × 125 A chiller
- CDA: > 8 bar
- Cooling capacity: 100 kW
- UPS: 24 VDC safe shutdown

14. Annex C – Control & Network Overview Diagram



15. Annex D – Standards Compliance Matrix

HyFleet compliance is structured against international, European, and industry-specific standards. Clause-by-clause evidence is held in the Certification Pack. This annex lists the applicable standards, aligned with the Intent, Requirements Matrix (RVM), and the supplied Standards Register.

CE/UKCA Directives (framework) :

- Machinery Directive 2006/42/EC
- Pressure Equipment Directive (PED) 2014/68/EU
- Electromagnetic Compatibility (EMC) Directive 2014/30/EU
- ATEX Directive 2014/34/EU

ATEX & Hazardous Areas (EN/IEC 60079 & related)

- BS EN 60079-0:2018+A11:2024 – Explosive atmospheres – Equipment – General requirements
- BS EN 60079-1:2014 – Protection by flameproof enclosures “d”
- BS EN IEC 60079-7:2015+A1:2018 – Increased safety “e”
- BS EN IEC 60079-11:2024 – Intrinsic safety “i”
- BS EN IEC 60079-10-1:2021 – Classification of areas – Explosive gas atmospheres
- BS EN 60079-13:2017 – Pressurised/ventilated rooms “p”/“v” (enclosure concept)
- BS EN 60079-14:2014 – Electrical installations – Design, selection, erection
- BS EN IEC 60079-15:2019 – Type of protection “n”
- BS EN IEC 60079-17:2024 – Electrical installations – Inspection & maintenance
- BS EN IEC 60079-19:2019 – Equipment repair/overhaul/reclamation
- BS EN IEC 60079-25:2022 – Intrinsically safe electrical systems
- BS EN IEC 60079-26:2024 – Equipment with separation elements / combined levels of protection
- BS EN 60079-29-1:2016+A11:2022 – Gas detectors – Performance requirements
- BS EN ISO 80079-36:2016 – Non-electrical equipment – Basic methods & requirements
- BS EN ISO 80079-37:2016 – Non-electrical protection (c/b/k)
- EN 1127-1:2019 – Explosion prevention and protection – Basic concepts

Hydrogen Fuelling / Gas Systems

- BS ISO 19880-1:2020 – Gaseous hydrogen – Fuelling stations – General requirements
- BS ISO 19880-2:2025 – Gaseous hydrogen – Dispensers & dispensing systems
- BS ISO 19880-3:2018 – Valves (selection guidance)
- BS ISO 19880-5:2019 – Dispenser hoses and hose assemblies

- BS ISO 19880-8:2024 – Fuel quality control
- BS ISO 19880-10 – Mobile/relocatable systems (if applicable)
- SAE J2601:2020 – Light-duty fuelling protocols
- SAE J2601-2:2020 – Medium & heavy-duty fuelling protocols
- SAE J2601-3 / J2601-5 – Additional fuelling topics (as applicable)
- SAE J2799 – Communication for H2 fuelling
- BS EN 17127:2024 – Outdoor hydrogen refuelling points
- EIGA Doc 211/17 – Hydrogen vent systems for customer applications
- EIGA CP-41 – ATEX separation distances on site
- EIGA CP-33 – Hydrogen system safety guidance
- BS EN 334:2005+A1:2009 – Gas pressure regulators (≤ 100 bar)
- BS EN 1349:2009 – Industrial process control valves

Machinery & Functional Safety

- BS EN ISO 12100:2010 – Safety of machinery – General principles
- BS EN ISO 13849-1:2023 – Safety-related parts of control systems – Design
- BS EN ISO 13849-2:2012 – Safety-related parts – Validation
- BS EN ISO 13850:2015 – Emergency stop function – Principles
- BS EN IEC 62061:2021 – Functional safety of safety-related control systems
- BS EN 60204-1:2018 – Electrical equipment of machines – General requirements
- BS EN ISO 4414:2010 – Pneumatic fluid power – Safety requirements
- BS EN ISO 4413:2010 – Hydraulic fluid power – Safety requirements
- BS EN 1012-1:2010 – Compressors & vacuum pumps – Safety requirements – Air compressors

Pressure Equipment / Piping (PED & design codes)

- EN 13480-1/-2:2024 – Metallic industrial piping – General / Materials
- BS EN 13445-1/-2/-5:2021(+A1:2023) – Unfired pressure vessels – General / Materials / Inspection & testing
- BS EN ISO 4126-1/-2/-7 – Safety devices for protection against excessive pressure (safety valves, bursting discs, common data)
- BS EN ISO 23251:2020 – Pressure-relieving and depressuring systems (aligns with API 521)
- PD CEN/TR 14549:2004 – Use of ISO 15649 & ASME B31.3 under PED
- BS EN ISO 9606-1:2017 – Qualification testing of welders – Steels

- BS EN ISO 9712:2022 – NDT personnel – Qualification & certification

Electrical / Low-Voltage Assemblies (applied under MD)

- BS EN IEC 60309:2022 (Parts 1,2,4) – Plugs, sockets, inlets – Industrial
- BS EN 60529:1992+A2:2013 – Degrees of protection (IP code)
- BS EN IEC 61439-1/-2/-7:2021 – Low-voltage switchgear & controlgear assemblies
- BS EN 61643-11:2012+A11:2018 – Surge protective devices
- BS EN IEC 60947-1/-2/-3/-5-1 – Low-voltage switchgear & controlgear
- BS EN 61010-1:2010+A1:2019 – Safety for measurement/control equipment
- BS EN 50288-1:2013 – Multi-element metallic cables – Generic spec
- BS 7671:2018+A3:2024 – Requirements for Electrical Installations (IET Wiring Regulations)

Structural / Mechanical & Lifting

- BS EN 1993-1-1:2022 / -1-5:2024 / -1-8:2024 – Eurocode 3 – Steel structures
- BS EN 10025 series / BS EN 10219 series – Structural steels & hollow sections
- BS EN ISO 10855-1:2024 – Offshore containers & lifting sets – Design/manufacture/markings
- BS EN ISO 13920:2023 – Welding – General tolerances
- ASTM A269 / A312 – Austenitic stainless tubing/pipe (selection)

Documentation, Diagrams & Labelling

- BS 8888:2020 – Technical product documentation & specification
- BS EN ISO 10628-1:2015 / -2:2012 – Diagrams for chemical/petrochemical industry

Specifications & symbols

- BS ISO 1219-2:2012 – Fluid power – Graphical symbols & circuit diagrams
- BS EN ISO 7010:2020+A8:2024 – Safety colours & registered safety signs
- BS EN 61882:2016 – HAZOP – Application guide

Cybersecurity, Quality & Management Systems

- BS EN IEC 62443-2-1:2024 / 62443-3-3:2019 – IACS security – Program & system requirements
- BS EN ISO/IEC 27001:2023+A1:2024 – Information security management systems
- BS EN ISO 9001:2015+A1:2024 – Quality management systems – Requirements
- BS EN ISO 45001:2023+A1:2024 – Occupational health & safety management systems
- BS EN ISO 14001:2015+A1:2024 – Environmental management system

Where hyphenated series or multiple parts are listed, applicability is as per the Standards Register and component manufacturer documentation. Items marked “applies to manufacturers” are used for selection and verification rather than in-house certification.

16. Annex E – Project Document

This annex lists the internal deliverables required for the HyFleet development project. Status tracking and version control are maintained in the separate **Document Register**.

- SDS – Software Design Specification
- ATEX Register (incl. HAC drawings, equipment certifications, inspection plan)
- Certification Pack (CE/UKCA evidence & compliance matrix)
- Risk Assessment & Safety Report (incl. SIL/PL determination)
- HAZOP report & LOPA report
- Requirements Matrix (RVM)
- Intent Document
- P&ID package
- Equipment Schedule
- Electrical SLD & detailed schematics
- Control architecture & network drawings
- Cabinet/JB layouts & device/IO schedules
- Mechanical detailed drawings & analyses (FEA, heat flow)
- Commissioning / FAT / SAT documentation
- SW validation & safety test documentation
- User & maintenance manuals
- Tech File (for CE/UKCA)